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THE LANDSCAPE OF *RUNX1* MUTATIONS IN BREAST CANCER AND ITS THERAPEUTIC POTENTIAL

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INTRODUCTION

Breast cancer remains the leading cause of female morbidity and mortality despite advances in screening, therapy and surveillance. Current therapeutic strategies aim to treat the disease after it has developed however side effects constantly present and cancer cells become resistant. Thus, there is a need to find alternative therapeutic strategies that target carcinogenic pathways with greater specificity and lower toxicity. The finding of *RUNX1* mutations in luminal breast cancer patients has put it in a spotlight as it is implicated in the disease development and progression. In view of this, there is a potential for RUNX1 to be a therapeutic target for breast cancer and this study has been conducted to understand the gravity of *RUNX1* mutations in breast cancer patients and its association with the disease progression.

METHOD

Meta-analysis was performed on 7823 breast cancer patients comprising 8393 profiled samples from 20 combined studies. Out of these patients, 243 express at least one *RUNX1* mutation.

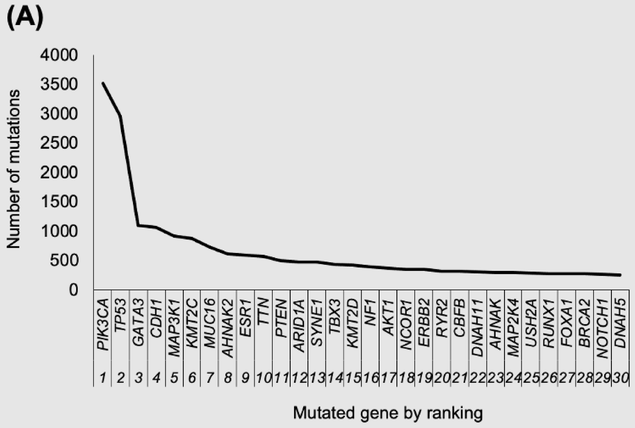
DISCUSSION

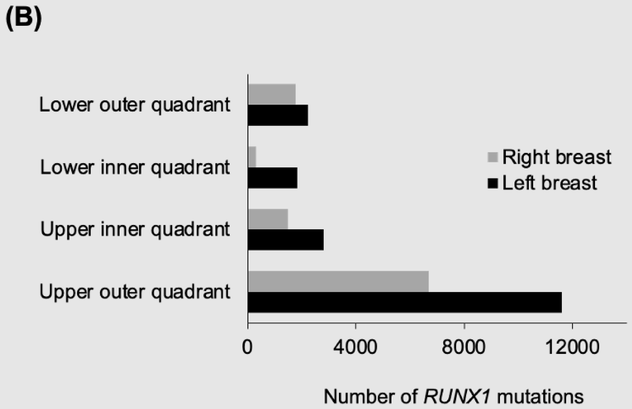
RUNX1 mutagenicity is prominent in breast cancer especially in women. The revelation that the left side of the breast has higher number of *RUNX1* mutation is intriguing as this may attribute to breast cancer pathogenesis especially when its mutation is highly detected in the early stage of the disease. However, a thorough analysis of *RUNX1* mutations is demanded in order to understand the consequence of these somatic changes in patients.

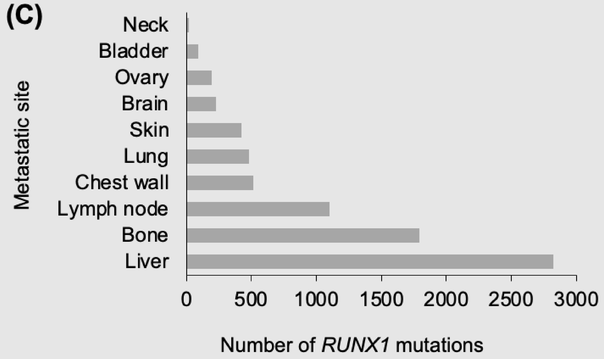
CONCLUSION

Unraveling *RUNX1* mutagenicity in breast cancer may provide insight into developing novel therapeutic strategies for the disease. This is important as it has become a challenge to treat breast cancer patients especially due to cancer resistance.

RESULTS







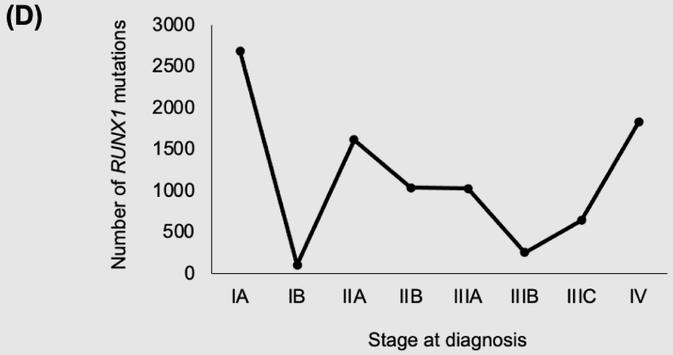


Figure 1: Meta-analysis of *RUNX1* mutations identified in 7823 breast cancer patients. (A) *RUNX1* is in the top 30 most mutated genes in breast cancer. (B) The left side of the breast has more *RUNX1* mutations than the right side. (C) The liver, bone and lymph node constitute the highest number of *RUNX1* mutations compared to other organs in the body. (D) Mutated *RUNX1* was detected at every stage of breast cancer.

Table 1: The great majority of *RUNX1* mutations are found in women compared to men with ER-positive breast cancer patients express 62% mutated *RUNX1* relative to ER-negative breast cancer patients that have only 38%.

Subject	*RUNX1* mutations (%)
Male	0.5
Female	99.5
ER-positive	62.0
ER-negative	38.0

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